Presentation 5
• Links to presentation(s) and code(s) on GitHub

Presentation:
https://docs.google.com/presentation/d/1ufGJ01XtTStk0VYXPpa5412CFFLEZTazgv_K8tP04Ks/edit?slide=id.g3aec66f797b_1_32#slide=id.g3aec66f797b_1_32

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/AllCalls_CAR_CallVolume_Dec8_Bonsitu_Kebeto.ipynb

Dashboard:
https://app.powerbi.com/links/ZhMneIecae?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare&bookmarkGuid=cec93178-6a6c-472c-9351-63ae8c45a4ba

https://app.powerbi.com/links/kui_n6E4m9?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare&bookmarkGuid=1e9c620b-48db-4963-a9df-df4cc888ae26

• What did you do?
I fixed the issue in my last presentation where some numbers like Trafficking Survivor were showing up as having no calls. I created two plots. Page 1 uses All Calls data to show which phone numbers people dialed (intake lines), while Page 2 uses CAR data to show which service categories callers reached (Family, Housing, Benefits, etc.).

• How does it help the project?
Now that the plots are fixed, we can accurately compare intake line volumes and service category demand over time. The two-page dashboard gives LegalAid stakeholders both perspectives: which phone numbers are receiving calls (for consolidation decisions) and which legal service areas have the most demand (for resource allocation).

• Issues faced (if any)

• Attempts to resolve issues (if any)

• Issues resolved (if any)

• Next steps
Continue to plot the rate of abandoned calls over the next year to see if any feedback used is leading to an overall decrease in the rate of abandoned calls.
Ask Cynthia what the numbers that end with the extensions 6023, 6342, and 6348 correspond with because they were the most commonly used unknown numbers.
Determine why there are over 300 unique numbers and if they are actually in use.

• References (Mention if you built up on someone else's work)
Used Muse's work for the inbound call classification logic (PSTN vendor name == "CallTower" and Direction == "TERMINATING"), using Correlation ID for unique call identification, and the first-leg approach for determining which intake line was dialed.

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Presentation 4

• Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_19Nov_Muse.pdf

Code:
https://app.powerbi.com/links/BcS5aPih9o?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare&bookmarkGuid=b3b5e62d-e08c-4c3f-9e7d-07f8e4d33811

Dashboard:

https://app.powerbi.com/links/BcS5aPih9o?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare&bookmarkGuid=dc493d61-3dbc-4fea-a917-c2ac1fc9aa24

• What did you do?
I fixed the issue in my last presentation where the aggregate bar chart correctly showed the Main number as having the highest volume but the monthly and quarterly charts incorrectly displayed Immigration as the highest-volume line. This issue stemmed from the code, which calculated a single aggregate average for all numbers combined within each time period. This aggregate average was then applied to every row within that period, regardless of the specific intake number. To fix this, I modified the code to group data by both time period and the number in the first leg of the call, ensuring that separate averages were calculated for each line. Additionally, I implemented a "Top 5" slicer in Power BI to limit views to the five highest-volume lines, improving data visualization and clarity.

• How does it help the project?
Now that the averages are calculated correctly, we can accurately compare the lines/

• Issues faced (if any)

• Attempts to resolve issues (if any)

• Issues resolved (if any)

• Next steps

More analysis for the spike in August for Internal VM, and in December for CLASP. Also, investigating why there seems to be one call for the Trafficking Survivor line. But, this is also probably because the chart shows the first leg of the call, but for the next presentation I could plot all transfers (at most 4) rather than the intake and first transfer only.

• References (Mention if you built up on someone else's work)

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Presentation 3
• Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_6Nov_Jasmine.pdf

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/Call_Volume_Nov6_Bonsitu.ipynb

Dashboard:

https://app.powerbi.com/links/VpAkdjldNu?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare

• What did you do?

Analyzed incoming call patterns for Legal Aid Chicago's 25 phone lines from October 2024 to September 2025. I tracked complete call journeys as callers got transferred between different numbers, using "Correlation ID" to count each unique call only once. I labeled each phone number with what it actually is (like "Main number" or "Bankruptcy Helpdesk") to make the data easier to understand. Then I calculated "Average Calls per Weekday" for different time periods (by hour, day, month, and quarter) by counting unique calls and dividing by how many weekdays were in that time period to have a better comparison.

• How does it help the project?
The analysis shows when and where calls are coming in, which helps Legal Aid figure out how to staff their phone lines. The "Average Calls per Weekday" metric makes it possible to compare busy and slow months without worrying about whether a month had more or fewer business days. The Power BI dashboard lets people drill down to see call volumes by time of day, day of the week, and specific phone lines. This helps identify peak call times and which lines aren't being used much, so Legal Aid can adjust staffing schedules and possibly reorganize their phone menu.

• Issues faced (if any)
The bars do not align on power BI with the x-axis labels, they're shifted one over, however I wasn't able to figure out what caused this or fix it.

• Attempts to resolve issues (if any)
I reloaded the csv and recreated the visual, tried different sort axis options (ascending/descending).

• Issues resolved (if any)
N/A

• Next steps
More analysis is needed of the volume of incoming calls per week because many of the different numbers have overlapping distributions, which seem too similar and further analysis into the relationship between monthly and quarterly trends.

• References (Mention if you built up on someone else's work)
Worked with Muse and used some of her previous preprocessing work.

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Presentation 2
• Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_21Oct_Jasmine.pdf

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/Bonsitu_FamilyMenuDrillthrough_Oct27.ipynb

Dashboard:
https://app.powerbi.com/links/kVcywmr7eL?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare

• What did you do?

Analyzed Family Menu navigation patterns by identifying all contact sessions that accessed the Legal Family Menu and categorizing them into three submenus: "Divorce or Parenting," "Education," and "General Family Issues." I counted unique sessions per category using "Contact Session ID". Created a 2-level hierarchy structure (Level 1: Family Menu aggregate; Level 2: three navigation categories) and aggregated monthly call volumes from October 2024 - September 2025 which shows how many unique sessions accessed each Family Menu category by month across the 2024-2025 period.

• How does it help the project?
The analysis enables Power BI drill-through functionality that allows stakeholders to view total Family Menu calls by month, then drill into specific months to see which of the three navigation paths (Divorce/Parenting, Education, or General Family Issues) callers are selecting. This reveals navigation patterns and helps identify which subcategories are most/least utilized, supporting menu optimization.

• Issues faced (if any)
N/A

• Attempts to resolve issues (if any)

N/A

• Issues resolved (if any)
N/A

• Next steps
Create a drill-through for the other sub-menus in legal menu and create a threshold to eliminate some categories..

• References (Mention if you built up on someone else's work)

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Presentation 1
• Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_9Oct_Jasmine.pdf

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/Bonsitu_Kebeto_Presentation1_Code.ipynb

Dashboard:
https://app.powerbi.com/links/l-JSupMipK?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare&bookmarkGuid=06c57976-3112-472b-b5b4-964a396b5eba
https://app.powerbi.com/links/goOOGrYVjf?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare

• What did you do?
I created two graphs, the first showing how call volume varied over time across intake lines. I did this by grouping the CAR dataset by month and intake line, and plotting the unique calls using "Contact Session ID".
The second graph looks at the peak call hours for the "Main Number Telephony EP" intake line, by first filtering the dataset for that specific intake line and grouping by hour, then counting the unique calls using "Contact Session ID".
I then exported the results to a csv and used that to create visualizations for both in Power BI.

• How does it help the project?
The first graph was to identify the intake lines with the most and least amount of traffic so that Legal Aid could remove the options that get the least amount of traffic to make intake simpler for the caller and combine it with other intake lines.

The second graph was in order to see which hours were busiest for their most used intake line so that Legal Aid could prioritize staffing during those hours.

• Issues faced (if any)
There were some questions I had to clarify about the difference between intake lines and the 6 numbers that customers access legal aid from.

• Attempts to resolve issues (if any)
I was able to clarify the questions I have with the documentation on Github and revise my recommendations.

• Issues resolved (if any)
I updated the presentation slides to be more specific about consolidating all of the various Legal Issues intakes lines into just one.

• Next steps
I want to use the All Calls dataset to calculate the duration of the calls and abandonment rate for each intake line to see the performance across different intake lines in order to further justify which specific lines should be combined or discontinued.

• References (Mention if you built up on someone else's work)

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